

Defending Against Identity Threats using Adaptive Authentication

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Abstract—Defending against identity-based threats and attacks which have tremendously grown in number in the age of remote working and access, requires intelligent, strategic, non-conventional, and dynamic ways of authentication and authorization. This paper aims to make identity security risk-based and hence adaptive by devising risk-scoring algorithms for five real-time use cases in detail. Zero-trust security principles are incorporated by continually collecting sign-in logs and analyzing them to check for any suspicious activities or anomalies to make it a dynamic approach. Based on the risk scores calculated users are segregated as risky and non-risky. While many adaptive authentication approaches have been proposed, the identities are confined just to users. Moreover, they lack emphasis on practical risk evaluation techniques. This work considers devices as an identity too and categorizes them as registered and unregistered devices. Further, results are made available to security administrators by displaying them on a dashboard for them to analyze and make necessary decisions like mitigation, multi-level authentication, or any other access control policies as such.

Keywords—Identities, zero trust principles, Risk-based authentication, Health-posture, Adaptive, Dynamic access control

I. INTRODUCTION

In today's cloud and remote working era, users access a significant number of corporate IT assets (business and enterprise applications, servers) via laptops or mobile devices to gain access (including privileged access) from outside the corporate-governed network boundary. This change in paradigm has resulted in many security challenges for firms, one of the biggest being "Identity-based threats" and exploiting these has led to many breaches. Due to externally exposed apps, user devices, emails, and identities, the attack surface has increased significantly. Granting access and trust based on static access control policies (which provide the same result in all situations and times) is still a big risk. There is a need for dynamic solutions where the changing environment and context are also considered.

The traditionally adapted, single-level authentication identified users solely based on their usernames and passwords.

Although multi-factor authentication has emerged, it is still not widely used. Single-factor authentication can't guarantee that users will refrain from malicious activities once authenticated or that they will always use compliant devices to login. Identities are not confined just to users, but devices also have their respective identities, which is their device health posture (physical address, OS, registered devices, and many other parameters could be considered). If device identity is neglected, then in today's age of remote access, where employees work from outside the company-bounded network, it wouldn't take even minutes for any malware to spread to other devices in the network, allowing attackers to escalate their privileges. Hence, conventional security measures have proven to be weak time and again [11,12,13] and there is a need for security measures that are dynamic and adaptive to the context. When it comes to authentication, many assumptions are made about the users; it would be ideal if the newly emerging zero-trust security model, which emphasizes explicitly verifying, giving least-privileged access, always assuming a breach, and trusting none, were incorporated.

A set of characteristics that uniquely identifies a person or thing is called an identity. User identities, such as usernames and passwords, could be used to authenticate the user. An identity could belong to an individual, an application, a device, or any other entity. Identity security offers a comprehensive solution for safeguarding every identity within an organization. This means that any identity, irrespective of their position or role, could escalate their privileges, leading to an attack path against the most valuable assets of the organization. As a result, privilege access management, which protects all identities, human or machine, is a critical identity security strategy throughout the critical access cycle.

Identity security strategy necessitates the most accurate authentication of identities, authorization with the appropriate permissions, and providing identities with structured access to assets of great privilege. To preserve the integrity of the

entire process, all these actions must be carried out in a way that is auditable (or accountable for). Instead of having to choose between security and productivity, identity security should allow organizations to secure access across any device, anywhere, and at the appropriate time. Treating identity as the first line of defense can potentially save a lot of time and effort.

Organizations have always had the need to protect themselves from identity-based threats. Over the last few years, the quantity and diversity of identities used have increased. Moreover, businesses have made a shift to cloud-based technology to provide consumers with an interactive digital experience. An increase in DDoS attacks has also been noticed due to the increase in the popularity of cloud and IoT-based systems [9]. Developments in technology have resulted in an increase in the remote workforce. Owing to the pandemic, there has been a rapid increase in the use of these services by businesses in 2020. However, attackers are also inventing new strategies like phishing, infected websites, social media (impersonation of accounts), and scanning and attacking internet-accessible areas of the company network [8]. Moreover, threats have become sophisticated over the years and have become difficult to identify [14]. This has led to expanded levels of danger.

A single hacked account is sufficient to start a security crisis. Attackers can escalate privileges or gather information once they've gained access, enabling them to accomplish their objectives. Identification has now become a security perimeter, enabling a decrease in data breaches and making it difficult for hackers to steal identities. Stronger protection systems are therefore required to protect identities. Authentication is the process of verifying that a person is who they claim to be.

Risk-based authentication aims to authenticate credentials based on the perceived risk of the requested connections or authorizations. While strict authentication is enforced when necessary, reducing authentication load improves the user experience. Unlike the static approach of verifying usernames and passwords, the dynamicity of RBA considers the authentication context and risk of the situation. This results in different outcomes based on the situation, making it an adaptive approach.

II. RELATED WORK

Owing to the flaws in non-continuous authentication in web applications, the authors of [1] suggested a risk-based authenticator and dynamic risk engines as a fix. One of its main advantages was that the user's authenticity was verified frequently (dynamically) and was transparent to the user. Despite being adaptive, the lack of focus on internal threats and device health were major flaws in this paper.

The authors of [2] present a systematic review and a comprehensive interpretation of modern dynamic access control models. The paper summarises and analyses in detail the risk factors and risk estimation techniques that can be considered for risk-based access control. Being context and risk aware, makes the methodology specific to the threat examined. No practical implementation has been provided in this paper to

back up the study. The focus is on attribute-based access controls, which are largely subject- and environment-based. This is ideal in terms of authorization and access control but complicated to gather, maintain and leverage numerous attributes.

In [3], the constantly changing modes of cyber attacks and ways to counter them were elucidated. The report placed emphasis on "zero-trust principles". The report also shed light on the zero-trust controls on the six pillars: network, applications, identities, infrastructure, data, and endpoints. It also presented statistics on attack categories like breach replay, password spray, and phishing that took place on compromised users. It also brings to light the increase in phishing emails. As per the report, cloud-based security is superior to on-premises security. However, zero-trust controls for the on-premises landscape and legacy systems were not addressed adequately.

The authors in [4] analyze risk-based authentication on eight platforms like Twitch, Amazon, LinkedIn, Google, and Facebook based on display resolution, IP address, time parameter, language, and user agent string. The platforms are classified as single-feature models, multi-feature models, and VIP models. Accounts are created on these platforms, and the behavior of these platforms is observed in different scenarios. The analysis was done to a great extent by creating over 200 accounts for inspection purposes, and the services were trained for a period of 3.5 months. This work did not consider all RBA approaches and also fixed the risk scores of the various situations beforehand, making the approach static.

The authors in [5] present an insightful risk-based authentication technique based on temporal access patterns for mobile devices and applications. Logs were gathered based on the user's access to the mobile applications, and features extracted from them were trained using random forest classifiers. This approach is employed in the background, making the approach dynamic and implicit (which reduces user intervention). The major drawback of this approach is that it uses the frequency of use of applications for classification and then provides authentication based on the usage. This is not a reliable indicator since many vital applications, like banking applications, contain sensitive user information and require a robust authentication method; however, they may not be used frequently. Moreover, this approach has not been tested on smartphone devices.

The authors in [6] provide an analysis of the dependability of system assets and user roles to improve the flexibility of adaptive MFA (A-MFA) systems with the help of a dynamic risk engine and risk-based authentication. The paper aims to propose a methodology to improve the robustness of A-MFA currently available. A methodology for calculating user trustworthiness and comprehending situations in which it arises is devised. It is observed that there are inaccuracies and false rejections in biometric technology. Moreover, the load on users using this approach for A-MFA was not measured.

Multiple re-authentication methods like Link (LINK), State of the Art (SOTA), and Subject (SUBJ) were performed inde-

pendently on volunteers and compared in [7]. A comparison of different attacks on these security solutions is performed in the presence of several users, which enables observation and detection of non-functional security issues. An analysis of the effectiveness of the security methods was backed by a statistical evaluation method. The test was carried out on a sample that doesn't necessarily represent the entire population. Re-authentication is often disabled due to it being prompted frequently. Thus, there's a possibility that statistical evidence has overlooked the emotional likelihood.

It is noticed that the majority of the papers do not make use of "zero trust principles," they lack a detailed evaluation technique for risk scores, and the emphasis is mostly on the risk factors alone. But how they contribute mathematically or in any other way to the risk score calculation is not explained. Most of the work considered "risk history" as a primary risk factor. Another observation is the negligence of the device as an identity, which is also important as the device a specific user uses to login also matters. Moreover, risk factors were application-specific and varied from paper to paper.

III. PROPOSED METHODOLOGY

The proposed methodology involves a dynamic form of risk analysis that detects anomalies, i.e., any form of suspicious behavior.

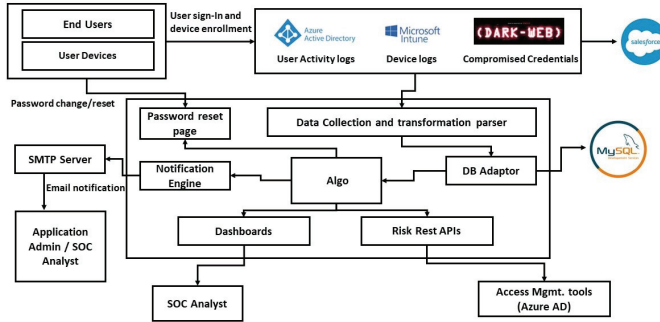


Fig. 1. Proposed Architecture

Fig. 1 demonstrates the proposed architecture which is a three-tier architecture where the log information is stored in a database in the backend, and the middle layer involves processing input and analyzing it using the devised algorithms to evaluate the risk scores and display results on the dashboard in the front end. The top layer includes information gathering, which is collected when end users interact with the Azure portal or Salesforce applications. All their sign-in logs are collected using MS Azure Active Directory, and the device-Id of the device from which users sign in can be captured using MS Intune. In general, compromised credentials are purchased from the dark web, but for implementation purposes, the credentials are hard-coded into a table in the database. Whenever a user logs in, their credentials are compared in the hardcoded table, and if there is a match, this implies there is a high risk associated with these credentials. The middle layer involves processing the gathered information. The "Data

collection and transformation parser" is used to parse JSON information (collected information would be in JSON format) into strings, which are then stored in the "MySQL" database using the "DB adaptor." Risk-scoring algorithms are used to analyze the stored information for different use cases. If further analysis is needed, then REST API calls can be made to Active Directory. After the analysis, dashboards and emails can be used to notify the application admins and SOC analysts.

This methodology has been implemented on a subset of real-life use cases (risk factors) such as:

A. Impractical Travel

This risk factor makes use of user travel velocity based on the login time and location and associates a travel risk score by comparing the velocity to the average speed of a flight [10]. This detects an impossible travel time between the locations and increases the risk score accordingly. The risk score can have any value between 0 and 5. Algorithm 1 describes the formulated algorithm for impractical travel.

Algorithm 1 Impractical Travel

Input: Geo points and time stamps of two logins

Output: Travel risk based on practicality

- 1: Calculate *speed* of travel between geo points of the two logins
- 2: **if** (Geo points are identical) **then**
- 3: $riskScore = 0$
- 4: **return** *riskScore*
- 5: **end if**
- 6: **if** (Identical time stamps and different geo locations) **then**
- 7: $riskScore = 5$
- 8: **return** *riskScore*
- 9: **end if**
- 10: $maxSpeed = 0.24$
- 11: $risk = 5 * (speed / maxSpeed)$
- 12: **return** *risk*

B. Abnormal Sign In

This risk factor derives a risk score based on abnormal sign-in logs by taking into consideration the user's very first sign-in attempt (during account creation), which gives the user's base or typical location, browser, and device, and the last 10 user sign-in logs. The risk score is calculated based on inconsistencies within the 10 sign-ins and with the initial sign-in. The risk score and confidence can be any number between 0 and 5. Algorithm 2 shows the calculation of location risk and confidence. The same algorithm is followed for device and browser risks as well.

C. Stolen Credentials

Often, stolen credentials are sold on the dark web. This use case compares the user's credentials upon sign-in and redirects them to the reset password page if they match the credentials in the compromised credentials table. Algorithm 3 shows the implementation of this use case.

Algorithm 2 Abnormal Sign In - Location

Input: User's location during account creation and last few user sign-in locations (country) from sign-in logs using Azure.

Output: Location Risk

Initialisation : Confidence = 0

```
1: if numLogs < 10 then
2:   total = numLogs
3: else
4:   total = 10
5: end if
6: for i = numLogs - total to numLogs - 1 do
7:   if location(i) = location(i + 1) then
8:     if location(i) = initialLocation then
9:       Confidence+=0.5
10:    else
11:      Confidence+=0.45
12:    end if
13:  else
14:    Confidence/=2
15:  end if
16: end for
17: locationRisk = 5 - Confidence
18: return locationRisk
```

Algorithm 3 Stolen Credentials

Input: Username and password

Output: Password risk

```
1: if Credentials found in compromised table then
2:   Redirect to reset password
3:   return 5
4: else
5:   Authenticate user credentials and redirect to login successful
6:   return 0
7: end if
```

D. Unregistered Device

Devices registered in Azure AD are considered safe. User sign-in logs are verified to ensure that a user logs in with a registered device. The sign-in logs with an empty "deviceId" field indicates a sign-in with an unregistered device. This information is collected using MS Intune. The dashboard displays whether or not the user logged in from a registered device.

E. Incognito Sign in

This risk factor verifies if the user has done any sign-in while hiding their identity by checking the "riskEventTypes" field of the user sign-in logs. This can be concealed by using the TOR browser to sign in, which results in user information being missing from the logs collected. This behavior is suspicious and requires action. Information on the user's anonymous login is displayed on the dashboard.

F. Deriving final risk score of user

The final risk score is calculated using a combination of all the calculated risk scores from the previously mentioned use cases. The risk scores are given weights based on importance and priority towards detection of identity threats, and a final risk score out of 5 is derived. The weights are assigned as a harmonic series. Table 1 presents the weights of the risk criteria.

TABLE I
DESCRIPTION AND WEIGHTS OF RISK SCORES

Risk Criteria	Description	Weight
Travel Risk	This holds the risk value derived from the impractical travel use case.	80/33
Location Risk	This holds the risk value derived from the abnormal location sign-in use case.	40/33
Browser Risk	This holds the risk value derived from the abnormal browser sign-in use case.	20/33
Device Risk	This holds the risk value derived from the abnormal device sign-in use case.	20/33
Password Risk	This holds the risk value derived from the stolen credentials use case.	5/33

$$TR = \frac{80}{33} * TravelRisk \quad (1)$$

$$LR = \frac{40}{33} * LocationRisk \quad (2)$$

$$BR = \frac{20}{33} * BrowserRisk \quad (3)$$

$$DR = \frac{20}{33} * DeviceRisk \quad (4)$$

$$PR = \frac{5}{33} * PasswordRisk \quad (5)$$

$$userRisk = (TR + LR + BR + DR + PR)/5 \quad (6)$$

IV. RESULTS AND DISCUSSIONS

Users were added to Azure Active Directory, and transactions were performed to generate sign-in logs for the users. These logs were collected using Java 11 with the help of graph APIs, analyzed, and then added to a MySQL database. Collection and analysis of logs were done every 30 minutes. Finally, the results were displayed on a dashboard using Java servlets to get data and bootstrap for the UI.

Sign-in logs of multiple users were collected using Azure Active Directory. The risk scores from the various use cases implemented are eventually combined to produce a final risk score.

Fig. 2 depicts the dashboard presented to the admin. The dashboard contains information about all the users, along with user examination and user mitigation options. Fig. 2 depicts the dashboard presented to the admin. The dashboard contains information about all the users, along with user examination and user mitigation options.

Fig. 3 depicts the user examine page displayed with the help of the "examine" button of the dashboard. It provides all

Admin Dashboard

Show 10 entries Search:

Username	Devices	Location	Number of Login Attempts	Confidence Level	Risk Level	Number of Violations	More Information	Action
admin	Linux	IN	336	4.69	0.31	0	Examine Mitigate	
chris	Windows 10	IN	70	4.98	0.02	0	Examine Mitigate	
Colby	Windows 10	IN	36	4.92	0.08	0	Examine Mitigate	
Alejandro	Windows 10	IN	15	1.9	3.1	1	Examine Mitigate	
Bonaly	Windows 10	IN	13	4.65	0.35	0	Examine Mitigate	
Brian	Windows 10	IN	12	1.76	3.24	1	Examine Mitigate	
Karthik	Linux	IN	12	4.51	0.49	0	Examine Mitigate	
Sarah	Windows 10	IN	12	1.74	3.26	2	Examine Mitigate	
Wilson	Linux	IN	12	4.3	0.7	1	Examine Mitigate	
Beatrice	Windows 10	IN	11	3.06	1.94	4	Examine Mitigate	

Showing 1 to 10 of 55 entries Previous 1 2 3 4 5 6 Next

Fig. 2. Admin Dashboard

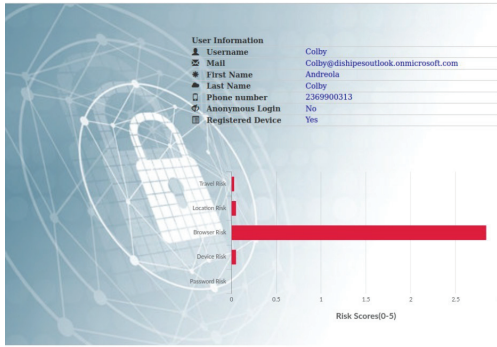


Fig. 3. User Information

user risk values and displays whether or not the user sign-in is anonymous or done using an unregistered device.

The "mitigate" button adds the user to the "Risk Investigation" group created in Azure Active Directory, as shown in Fig. 4. Further action can then be taken against these users.

Below are the logs and risk scores of three users to explain the working of the implementation.

A. User 1- Karthik

TABLE II
SIGN-IN LOGS OF KARTHIK

Date-time	Country	Browser	Device
User initial sign in			
2022-09-23 16:17:18	England	Edge 105.0.1343	Windows 10
Last 10 logs			
2022-09-23 16:20:25	England	Edge 105.0.1343	Windows 10
2022-09-23 16:20:16	Netherlands	Edge 105.0.1343	Windows 10
2022-09-23 16:19:30	Netherlands	Edge 105.0.1343	Windows 10
2022-09-23 16:18:47	Netherlands	Edge 105.0.1343	Windows 10
2022-09-23 16:18:30	Netherlands	Edge 105.0.1343	Windows 10
2022-09-23 16:18:20	Netherlands	Edge 105.0.1343	Windows 10
2022-09-23 16:17:52	Netherlands	Edge 105.0.1343	Windows 10
2022-09-23 16:17:49	Netherlands	Edge 105.0.1343	Windows 10
2022-09-23 16:17:27	Netherlands	Edge 105.0.1343	Windows 10
2022-09-23 16:17:18	Netherlands	Edge 105.0.1343	Windows 10

Search by name Add filters

Name	Type	Email	User type
☐ AS Akinkuolie Sarah	User		Member
☐ CA Chace Beatrice	User		Member
☐ CT Champaigne Brian	User		Member

Fig. 4. Risk Investigation Group

- 1) Travel Risk: The travel risk for Karthik was evaluated as 0. This conclusion is obvious because Karthik's last ten sign-in logs have all come from the same location, as shown in Table II.
- 2) Location Risk: The location risk of Karthik was found to be 0.5. Even though Karthik has a consistent location in the last 10 logs (Table II), they differ from his base location (the initial location on account creation), thus making the confidence level 4.5.
- 3) Browser Risk: The browser risk for Karthik was found to be 0.05. This is because, although there is no risk involved (the last 10 browsers match the initial browser as seen in Table II), a minimum risk of 0.05 is maintained.
- 4) Device Risk: The device risk for Karthik was evaluated at 0.05. This is because, although there is no risk involved (the last 10 devices match the initial device as seen in Table II), a minimum risk of 0.05 is maintained.

B. User 2- Alejandro

- 1) Travel Risk: The travel risk for Alejandro was found to be 5. It is observed from Table III that in the last 10 logs, the country kept changing between India and Denmark. Hence, the travel risk of 5 is justified. Because there has been a continuous change in locations with a very slight time difference, this could be a DDoS attack situation.
- 2) Location Risk: As calculated by the algorithm, it is found that the location risk for Alejandro is 3.42. This can be justified due to the pattern of the locations seen in the last 10 records in Table III. As shown in table III, there is a slight consistency in location between India and Denmark, but the location changes three times in the last ten sign-in logs. This gives rise to a high (> 3) location risk.
- 3) Browser Risk: Alejandro's browser risk is found to be 2.57. This value is justified since it is observed from Table III that the browsers, though mostly consistent, have changed twice in the last 10 sign-in logs, one of which matches the initial browser. This results in moderate browser risk.
- 4) Device Risk: The device risk for Alejandro was evaluated at 0.05. This is because, although there is no risk involved (the last 10 devices match the initial device as seen in Table III), a minimum risk of 0.05 is maintained.

C. User 3- Wilson

- 1) Travel Risk: The travel risk for Wilson was found to be 0.26. This can be justified due to the consistency displayed in the locations as well as the appropriate time

TABLE III
SIGN-IN LOGS OF ALEJANDRO

Date-time	Country	Browser	Device
User initial sign in			
2022-09-16 10:37:35	Poland	Firefox91.0	Windows 10
Last 10 logs			
2022-09-22 13:40:34	Denmark	Firefox 91.0	Windows 10
2022-09-22 13:39:07	India	Edge 105.0.1343	Windows 10
2022-09-22 13:39:04	India	Edge 105.0.1343	Windows 10
2022-09-22 13:38:02	India	Edge 105.0.1343	Windows 10
2022-09-22 13:37:55	India	Edge 105.0.1343	Windows 10
2022-09-22 13:37:28	Denmark	Firefox 91.0	Windows 10
2022-09-22 13:36:47	Denmark	Firefox 91.0	Windows 10
2022-09-22 13:36:08	Denmark	Firefox 91.0	Windows 10
2022-09-22 13:35:14	Denmark	Firefox 91.0	Windows 10
2022-09-16 10:37:35	Poland	Firefox 91.0	Windows 10

TABLE IV
SIGN-IN LOGS OF WILSON

Date-time	Country	Browser	Device
User initial sign in			
2022-09-09 09:34:32	India	Firefox 103.0	Linux
Last 10 logs			
2022-09-16 10:05:10	Luxembourg	Firefox 91.0	Windows 10
2022-09-16 10:05:10	Luxembourg	Firefox 91.0	Windows 10
2022-09-16 10:05:10	Luxembourg	Firefox 91.0	Windows 10
2022-09-09 09:35:32	India	Firefox 103.0	Linux
2022-09-09 09:35:21	India	Firefox 103.0	Linux
2022-09-09 09:35:14	India	Firefox 103.0	Linux
2022-09-09 09:35:14	India	Firefox 103.0	Linux
2022-09-09 09:34:32	India	Firefox 103.0	Linux
2022-09-09 09:34:32	India	Firefox 103.0	Linux

difference shown at the points where there is a change in location, as depicted in Table IV.

- 2) Location Risk: The location risk for Wilson was found to be 2.17. The change in locations with consistent intervals (Table IV) indicates that the user is moderately risky.
- 3) Browser Risk: Wilson's browser risk was found to be 2.17. From the logs in Table IV, it is observed that even though the signing-in has been done from the same browser, the version of the browser was different, hence attributing a value of 2.17 to the browser risk.
- 4) Device Risk: The device risk for Wilson was found to be 2.17. It can be observed in the logs (Table IV) that the devices used to login have been inconsistent (Windows 10 and Linux).

V. CONCLUSION

The paper proposed a modern, strategic, and innovative methodology of authentication that treats identity as a security parameter rather than a network. The collection and analysis of user logs every 30 minutes ensured the approach was dynamic. The zero-trust principles of "keep verifying and always assume breach" are also incorporated through this methodology. Risk-scoring algorithms were devised for real-time use cases based on the situation and context, making this approach adaptive. Hence, unlike traditional approaches, authentication is done beyond just the credentials by giving utmost priority to the risk scores obtained.

The scope of this paper is limited to a subset of real-time use cases but, more real-time situations can be explored. The proposed methodology considers users and devices as identities, but it can also be extended to service applications. This work considers device registration as a device health parameter however various other criteria like compliant operating system, malware free devices, unpatched settings can be considered. Although this paper introduced mitigation, the future scope could also incorporate more sophisticated actions and access control decisions.

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